

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 19

Title: Ketone body metabolism and Ketosis

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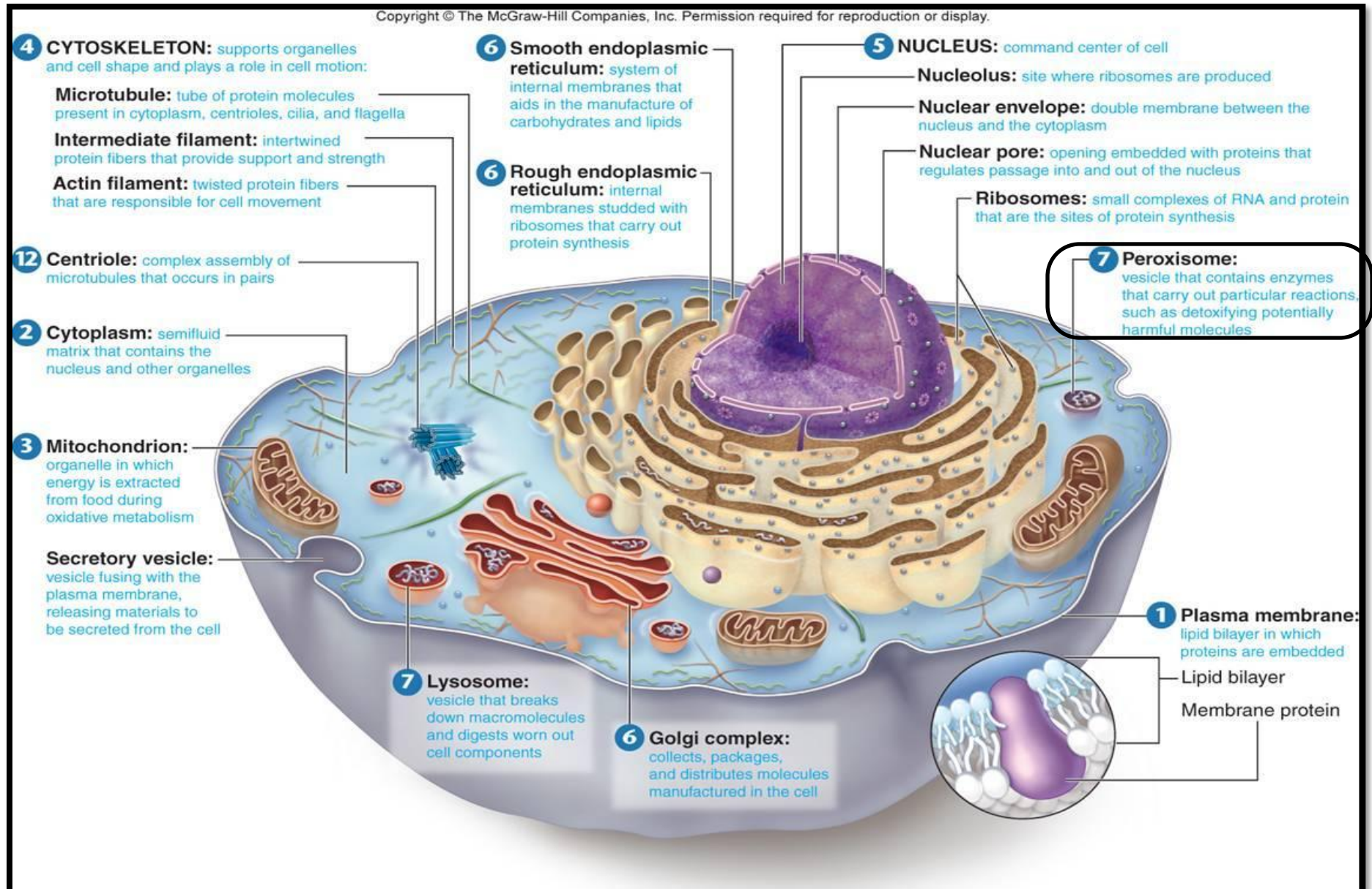
Objectives: Ketone body metabolism and disorders

SOM. MK.I.BPM2.3.DM.2.BCHM.1203	Outline peroxisomal oxidation and omega-oxidation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1204	Discuss Zellweger syndrome as a disorder of peroxisomes
SOM. MK.I.BPM2.3.DM.2.BCHM.1205	Describe hepatic ketogenesis highlighting significance of mitochondrial HMG CoA synthase. List the major ketone body produced during hepatic ketogenesis .
SOM. MK.I.BPM2.3.DM.2.BCHM.1206	Explain ketone body utilization in peripheral tissues and significance of thiophorase
SOM. MK.I.BPM2.3.DM.2.BCHM.1207	Prepare a concept map indicating the steps involved in the generation of ketosis in starvation and uncontrolled type 1 diabetes mellitus.
SOM. MK.I.BPM2.3.DM.2.BCHM.1208	Correlate laboratory data in ketoacidosis (laboratory data in blood and urine) to the clinical signs in a type 1 diabetic
SOM. MK.I.BPM2.3.DM.2.BCHM.1209	Explain the biochemical basis of occurrence of hypoglycemia and hypoketosis in the systemic fatty acid oxidation disorders

Recommended Reading

- Lippincott's Biochemistry 9th Edition: Chapter 16
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Reference: Lippincott's Biochemistry Lippincott's Chap 16,pg 189-200;
Concept map Fig 16.25; [Fatty Acid, Triacylglycerol, and Ketone Body Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- [Fatty Acid Oxidation Disorders: Physiology | Premium Basic Sciences | Health Library](#)
- Lippincott's Table Fig. 28.29 (Vitamins reference)
- **Lippincott's Questions: 16.1 – 16.5**

Peroxisomes and fatty acid oxidation



Peroxisomal oxidation of fatty acids (Zellweger syndrome)

- **Very** long chain fatty acids (22 to 26 C atoms) oxidized in peroxisomes
 - Shortened fatty acid next sent to mitochondria
- **Zellweger syndrome:** Defective peroxisomal biogenesis (liver and brain)
- **Increased C 22-26 fatty acids**
- Neurological manifestations (delayed development) and extensive demyelination
- Hepatomegaly and hepatocellular failure
- Usually fatal in infancy



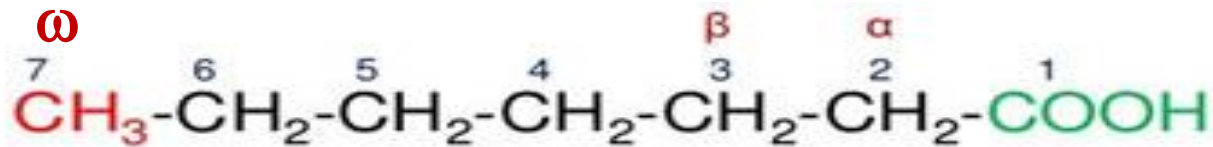
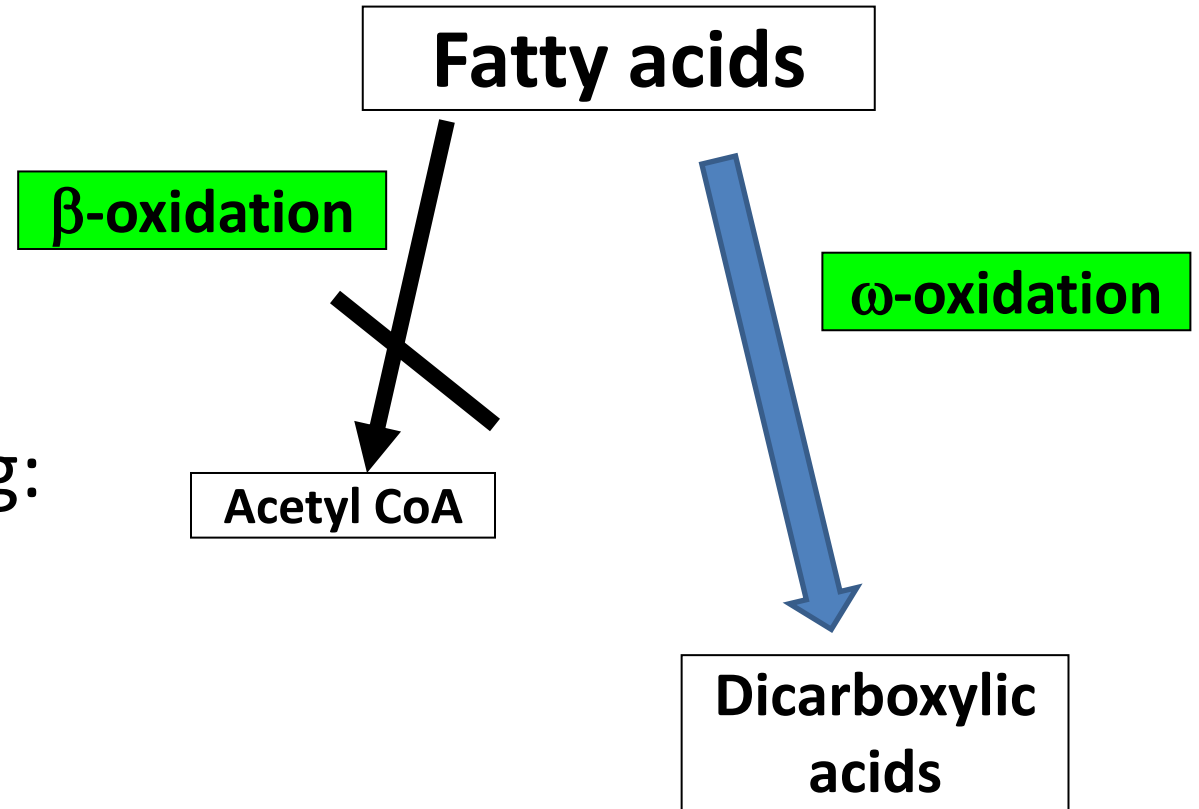
Alpha-oxidation of branched chain fatty acids (Refsum disease)

- **Phytanic acid (branched chain fatty acid)** in dairy products
- **α -oxidation** takes place in **peroxisomes**
- **Refsum disease:** Peroxisomal phytanyl CoA α -hydroxylase deficiency (α -oxidation defect)
 - Phytanate accumulates
 - Visual defects, ataxia and polyneuropathy and skeletal manifestations
 - Restrict **branched chain fatty acids (phytanic acid)**



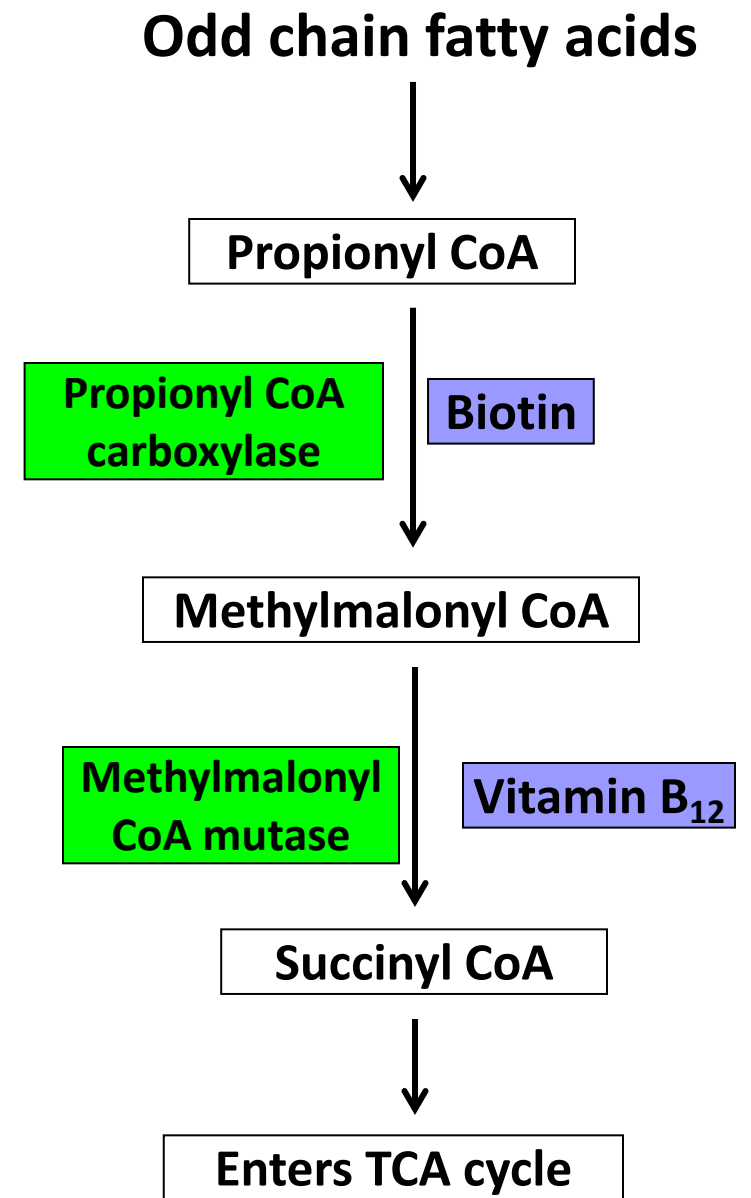
Omega (ω)-oxidation

- Minor pathway for fatty acid oxidation in ER
- Oxidation of ω -C atom of fatty acid, forms **dicarboxylic acid**
- When β -oxidation is defective (eg: MCAD deficiency), dicarboxylic acids found in urine



Oxidation of odd chain fatty acids

- Forms **propionyl CoA** (3C) in final round of β -oxidation
- Propionyl CoA $\rightarrow$ $\rightarrow$ Succinyl CoA (via propionyl CoA carboxylase and methylmalonyl CoA mutase)
- Propionyl CoA carboxylase needs ABC
 - ATP, **Biotin**, CO_2
- Succinyl CoA enters TCA cycle



Refer: Lippincott's Biochemistry 8th Edition: Table for vitamins
(other names; coenzyme forms; function): Pg: 396, 397

Summary table for fatty acid oxidation disorders

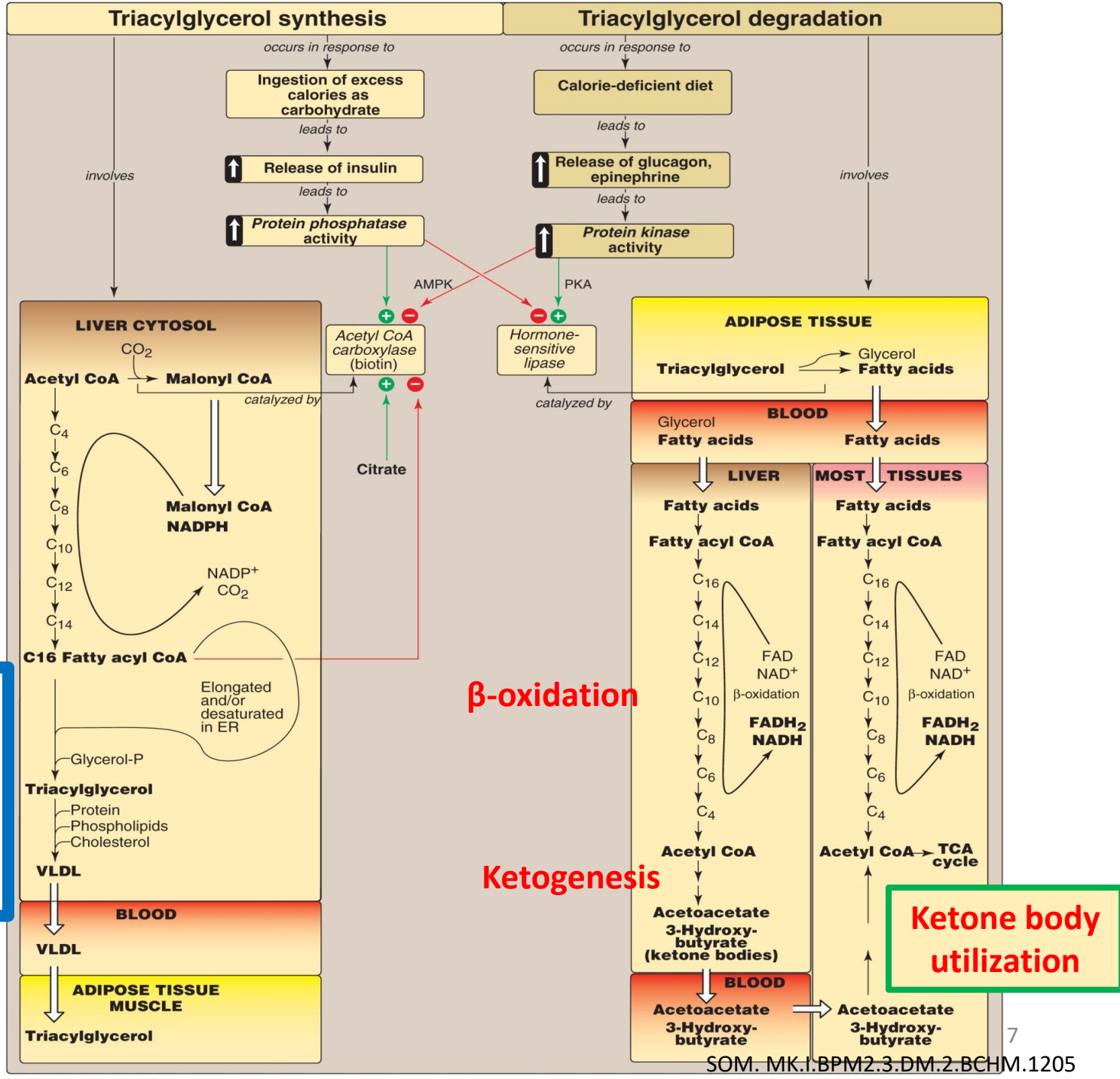
	Alpha oxidation	Very long chain fatty acid oxidation	Myopathic fatty acid oxidation defect	Systemic beta oxidation defect
Disorder				
Enzyme defect/ organelle involved				
Accumulating substrate				
Clinical features				
Lab findings				

Answers in Practice Questions

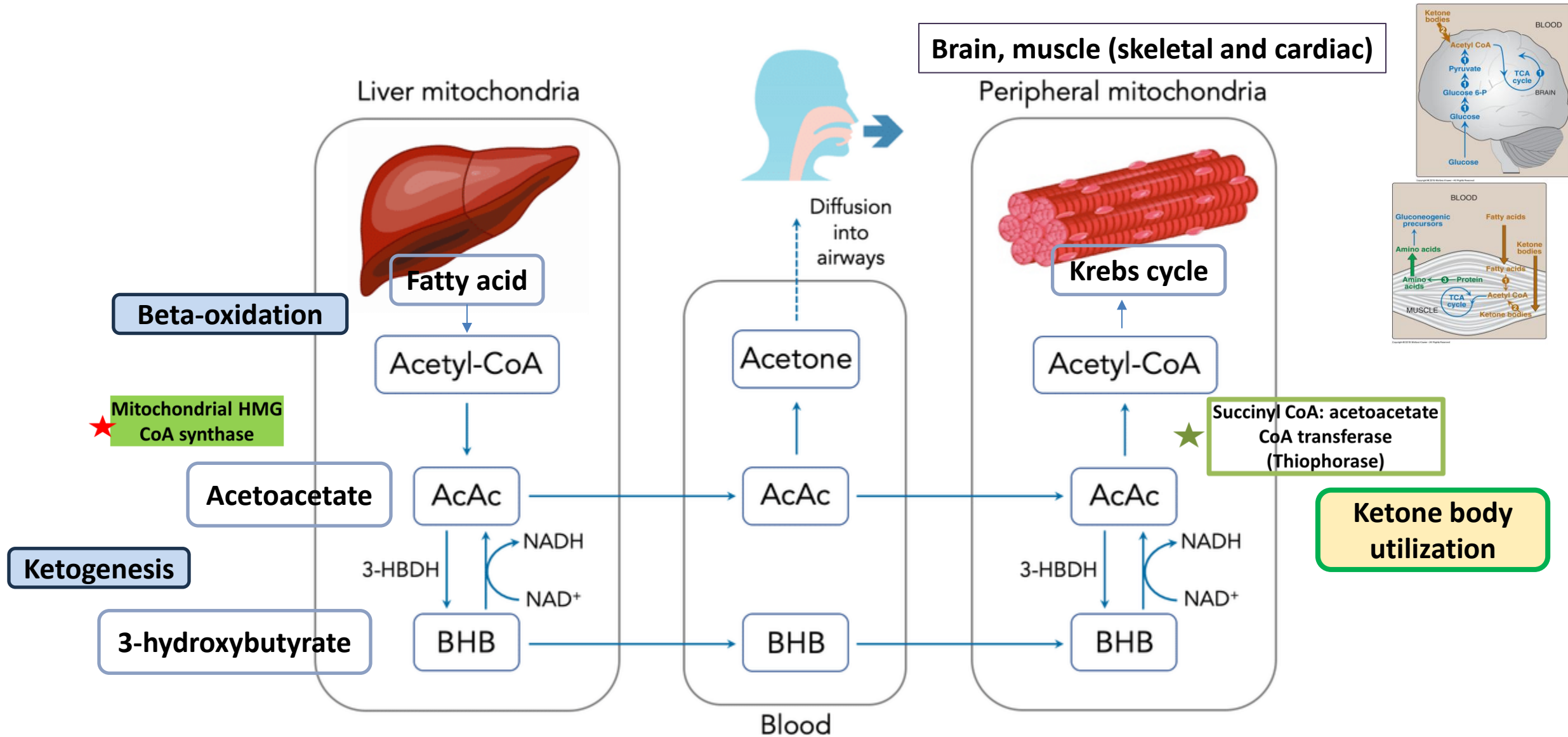
Purpose of pathway: Liver; Brain and muscle

Regulation: Fed vs fasting

Disorders



Overview of ketone body metabolism



Ketone bodies

- Acetyl CoA from fatty acid oxidation used to form ketone bodies in **liver** (peripheral tissues cannot synthesize ketone bodies)
- Ketone bodies: Acetoacetate, **3-hydroxybutyrate (β -hydroxybutyrate)** and acetone
- **Acetoacetate** and **3-hydroxybutyrate** transported to peripheral tissues
- Acetoacetate and 3-hydroxybutyrate **weak acids** (give off protons)
- In peripheral tissues reconverted to acetyl CoA $\rightarrow$ TCA cycle
- Ketogenic amino acids: Leucine and lysine $\rightarrow$ Form ketone bodies

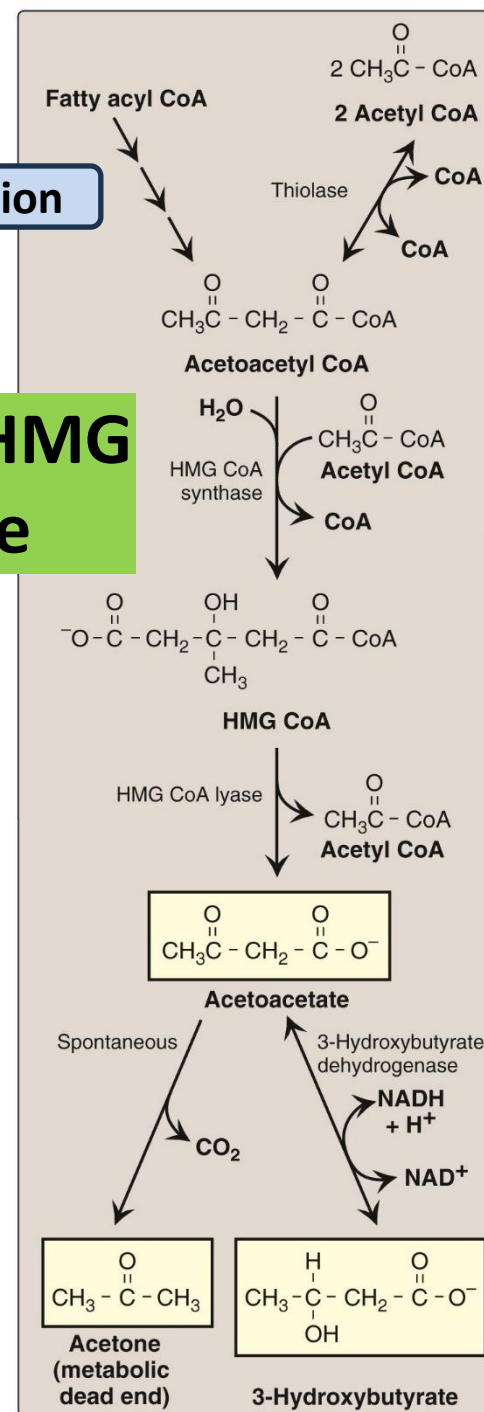
Ketogenesis

Regulated enzyme of ketogenesis

★ Mitochondrial HMG CoA synthase

Liver mitochondria

Beta-oxidation



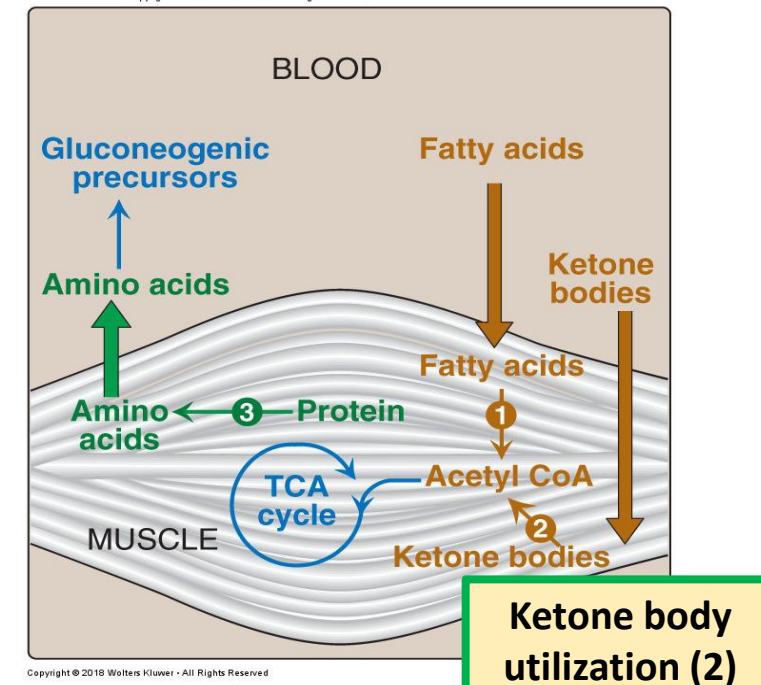
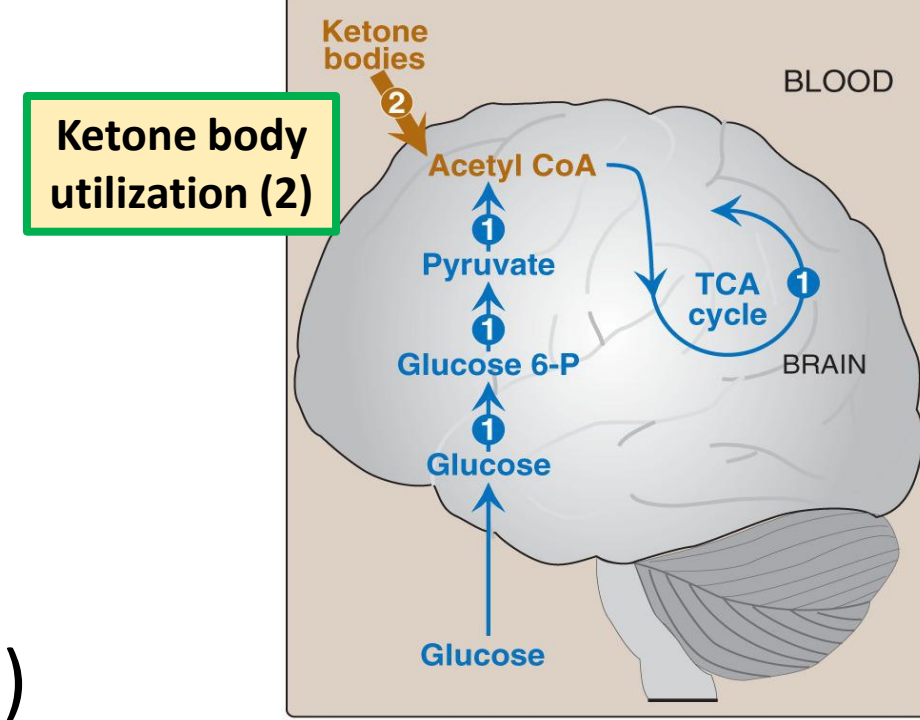
NADH/NAD⁺ is increased

Ketogenesis

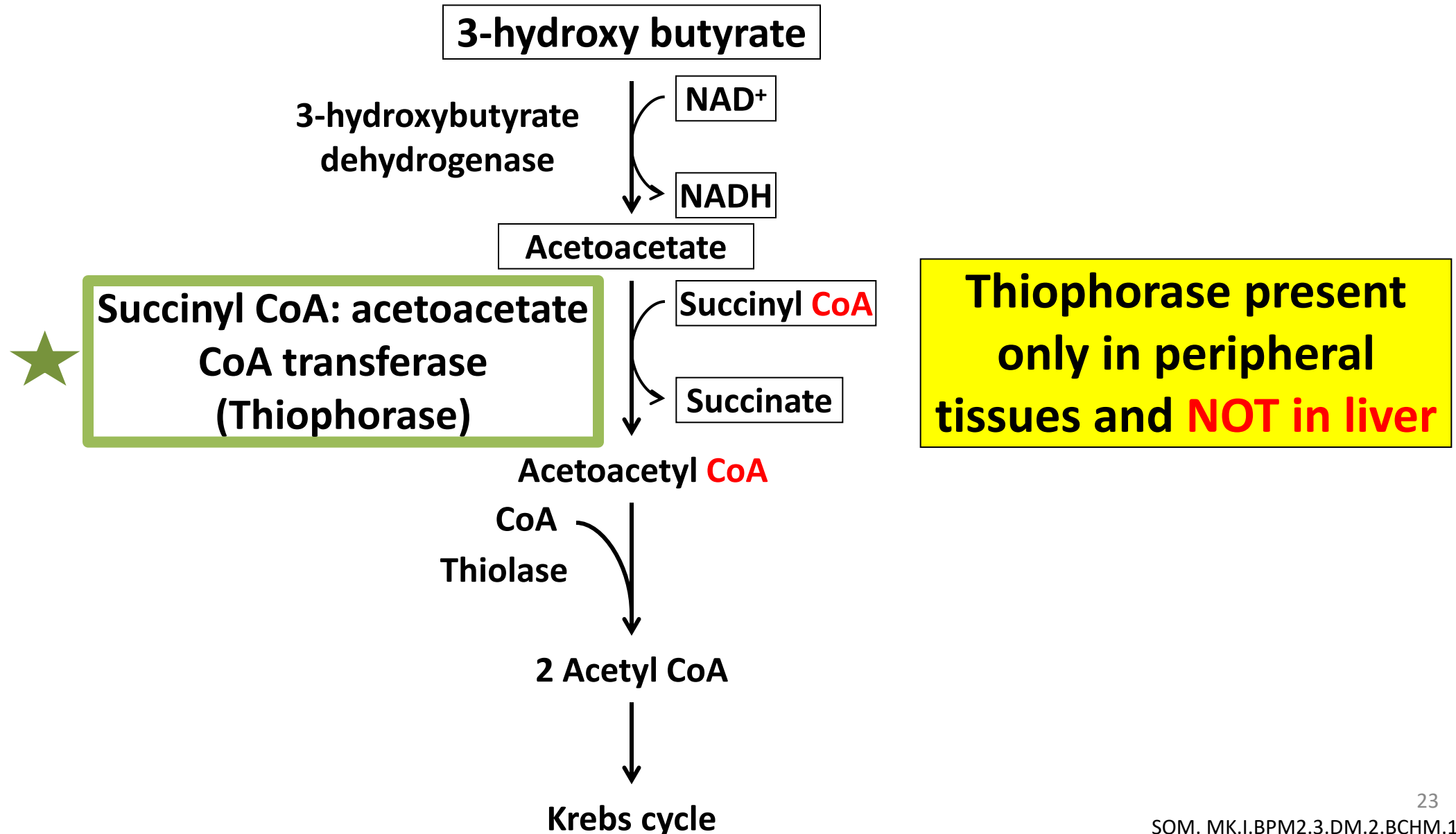
- Liver mitochondria
- **Mitochondrial HMG CoA synthase** – Rate limiting enzyme of ketogenesis (Differentiate from HMG CoA synthase of cholesterol synthesis) – **More active in starvation and type 1 diabetics with ketosis**
- **Spontaneous (non-enzymatic)** decarboxylation of acetoacetate forms **acetone** (volatile, **non-metabolizable compound**).
Acetone lost via lungs (fruity breath odor)
- **3-hydroxybutyrate** formed when increased NADH/NAD⁺ (Active β -oxidation)
- During fasting (ketosis), major ketone body **3-hydroxybutyrate**

Utilization of ketone bodies

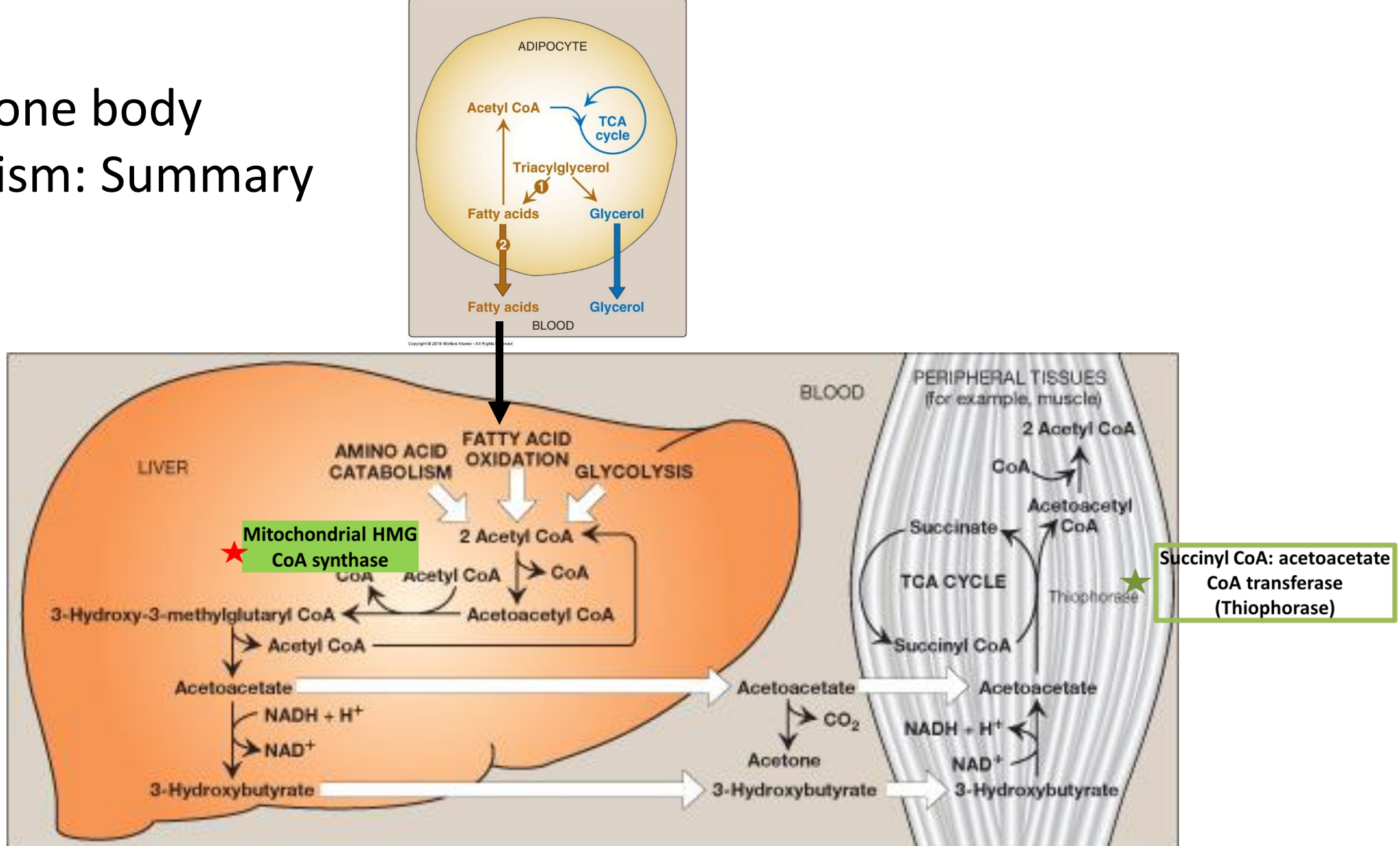
- Ketone bodies are formed in liver
- Used by peripheral tissues (**skeletal and cardiac muscle, brain**)
- Alternate fuel for **brain** (prolonged fasting)
- 3-hydroxybutyrate oxidized to acetoacetate
- **Succinyl CoA: acetoacetate CoA transferase (thiophorase)**
- Acetoacetyl CoA $\rightarrow$ Acetyl CoA $\rightarrow$ TCA cycle



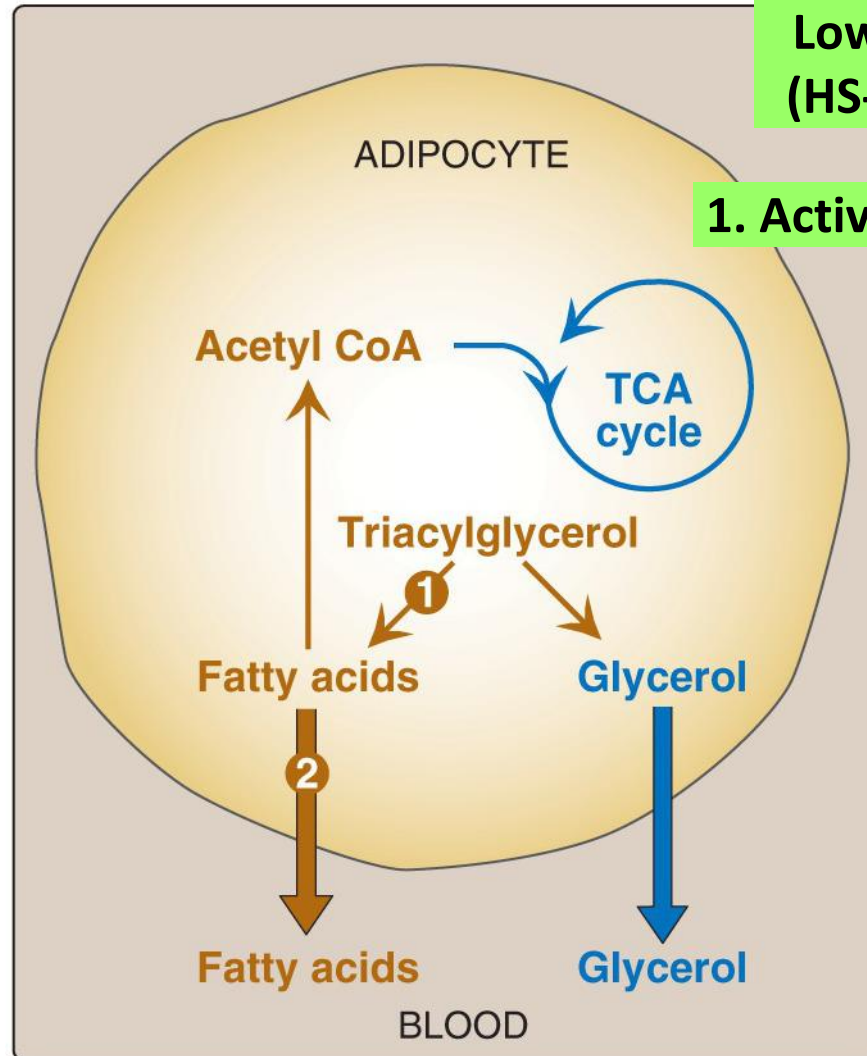
Utilization of ketone bodies



Ketone body metabolism: Summary

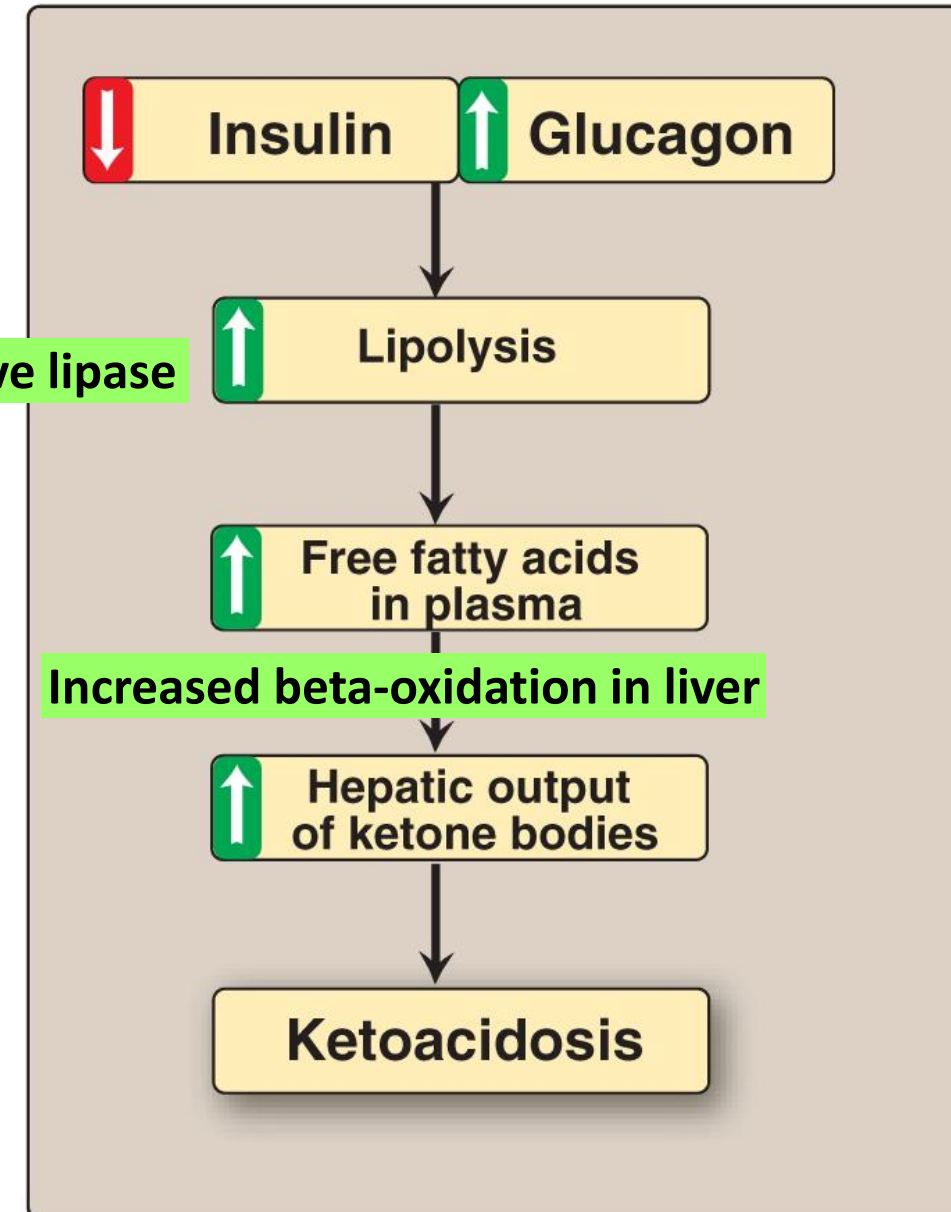


Mechanism of ketosis in starvation

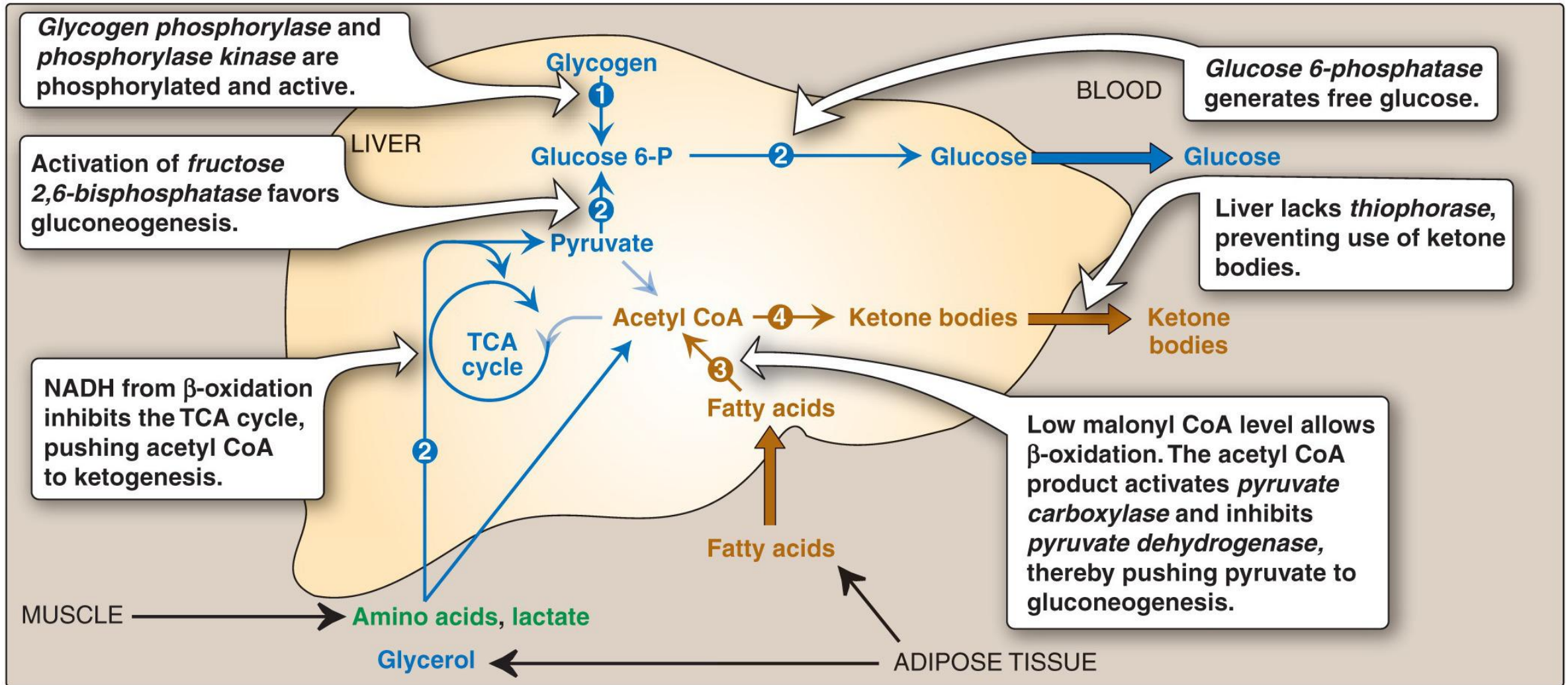


Low insulin/ glucagon ratio
(HS-Lipase phosphorylated)

1. Activation of hormone sensitive lipase



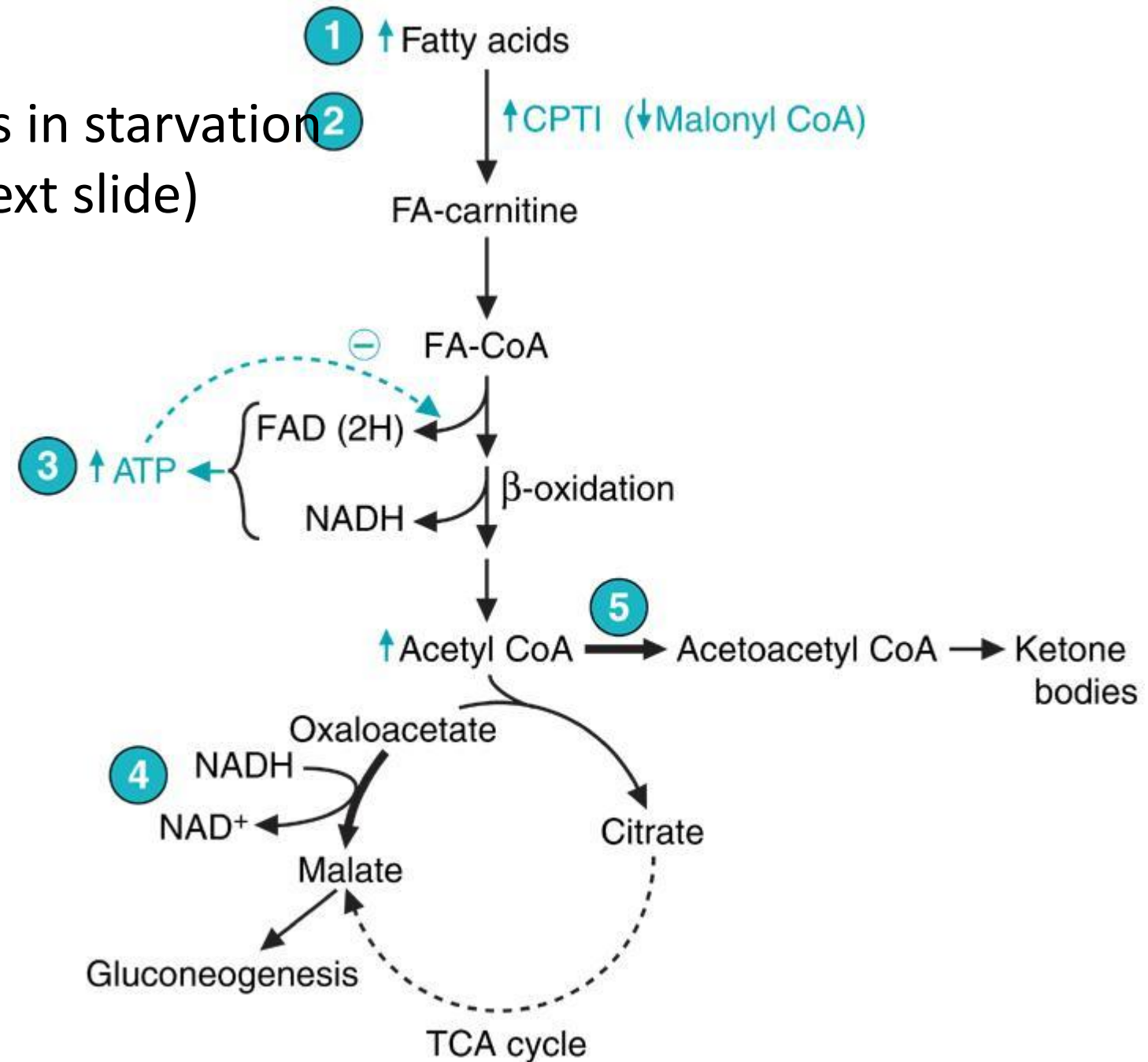
Mechanism of ketosis in starvation



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Ketone body Production > Use of ketone bodies by peripheral tissues = Ketosis

Mechanism of ketosis in starvation (explained on next slide)



(from Mark's textbook)

Mechanism of ketosis in starvation

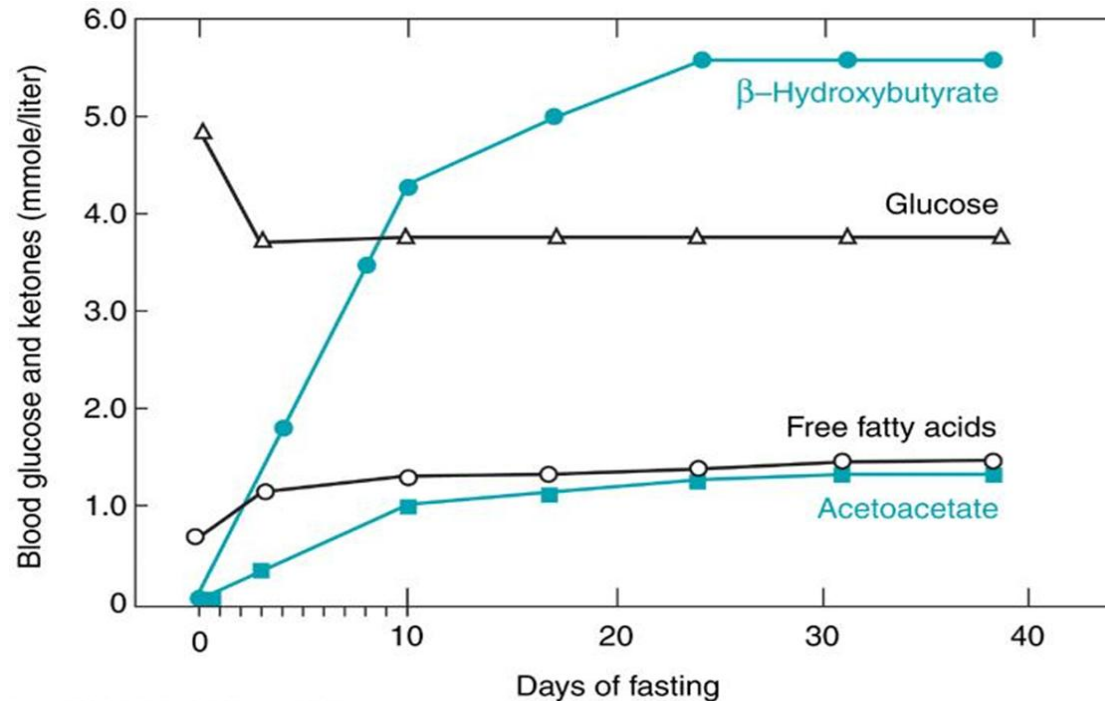
- Decreased insulin/ glucagon ratio
- Hormone sensitive lipase (phosphorylated) active
- Increased adipose tissue lipolysis
- Increased free fatty acids in circulation (1)
- Increased β -oxidation in liver (CPT-I active; Low Malonyl CoA) (2)
- β -oxidation increases NADH/ NAD⁺ and increases ATP levels (3)
- Increased acetyl CoA and increased ketogenesis (Active mitochondrial HMG CoA synthase)
- Acetyl CoA activates **pyruvate carboxylase** (gluconeogenesis)
- Pyruvate is shunted to gluconeogenesis
- Oxaloacetate for gluconeogenesis rather than Krebs cycle (4)
- Acetyl CoA towards ketogenesis rather than Krebs cycle (5)

Mechanism of ketosis in starvation

- Mild elevation in ketone bodies observed in 12-14 hr fast
- Ketoacidosis is complication in prolonged starvation
- Weak acids buffered by bicarbonate; pH decreases
- Fatty acids NOT a fuel for brain even during starvation
- Ketone bodies used by **brain** during starvation – Spares muscle protein breakdown (amino acids) for gluconeogenesis – '**Protein sparing effect** of ketone bodies'
- Remember: β -oxidation provides energy and acetyl CoA to activate gluconeogenesis. **BUT, C atoms of acetyl CoA (Fatty acids) NOT directly used as precursors of gluconeogenesis.**



Changes in metabolite levels in fasting



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Glucose: Reduced to 3.8 mM at day 3: Due to ketone body utilization by brain in prolonged fasting

Fatty acids: Increased and plateau after day 3: Due to Hormone-sensitive lipase (Lipolysis)

Acetoacetate: Increased: Due to adipose tissue lipolysis and hepatic ketogenesis

Hydroxybutyrate: Highly elevated (Major ketone body): Due to adipose tissue lipolysis and ketogenesis

Case report: Diabetic ketoacidosis

- A 27-year-old woman with 10-year history of type 1 diabetes mellitus brought in an obtunded state; She has nausea, vomiting and abdominal pain; Drowsy
- Dehydrated; blood pressure is low
- Breathing is deep and rapid (Kussmaul breathing)
- Fruity odor of breath (acetone)
- Plasma glucose: 400 mg/dL (Normal<125mg/dL)
- Arterial pH: 7.18 (Normal: 7.35- 7.45)
- Bicarbonate:12mEq/L (Normal:24-26)
- PCO₂=25mmHg (Normal:40)
- Anion gap: 28 (Normal: 8-14)
- Blood ketone levels: 15mM (Normal: 0.2mM on a normal diet)
- Urine positive for glucose and ketones



Ketoacidosis in uncontrolled diabetes mellitus

- Mechanism of ketogenesis in type 1 diabetic is similar to starvation
- Uncontrolled type 1 diabetes mellitus, excessive adipose tissue **lipolysis** (**very low insulin levels**) **Hormone sensitive lipase** very active
- Ketogenesis by liver (**active mitochondrial HMG CoA synthase**) is more than ketone body utilization by peripheral tissues → very high ketone bodies levels (**ketonemia**)
- Ketone bodies lost in urine (**ketonuria**) – detected by dipstick
- Weak acids and tend to lose protons
- Protons buffered by HCO_3^- , serum HCO_3^- levels fall, causing severe acidosis (metabolic acidosis)
- High anion gap metabolic acidosis (Ketone bodies are unmeasured anions)

Ketoacidosis in uncontrolled diabetes mellitus

- Elevated **3-hydroxybutyrate** and acetoacetate levels in blood and urine
- Acetone production increased
- “Fruity odor” of breath (Loss of volatile acetone)
- Increased rate and depth of respiration –hyperventilation (decreases PCO_2), compensatory response in metabolic acidosis
- Why is ketoacidosis more common in type-1 diabetes but not so common in type-2 diabetics??
- Differentiate between ketosis in prolonged starvation and ketoacidosis in type-1 diabetic??

Alcoholic ketoacidosis

- Observed in chronic alcohol use with malnutrition and recent **binge-drinking**
- Ethanol metabolism → Increases NADH/NAD⁺ ratio
- Ethanol → Acetaldehyde → Acetate → Acetyl CoA → Ketone body synthesis
 - 3-hydroxybutyrate >>Acetoacetate (due to increased NADH/NAD⁺ ratio)
- Maybe accompanied by
 - Hypoglycemia (Increased NADH/NAD⁺ ratio → Malate trapping)
 - Lactic acidosis

Case report: MCAD deficiency

- An 18-month-old girl brought to the emergency department in an unresponsive state.
- Episodes of hypoglycemia since she was 11 months old
- Occurred on fasting or **following illness**
- On admission (metabolic crisis),
 - **Serum glucose is 38mg/dL**
 - Free fatty acids in blood is elevated (normal), but
 - **Ketone bodies are below normal**
- Explain metabolic basis for hypoketonemia?

Hypoketonemia in systemic fatty acid oxidation disorders

- Defect in β -oxidation, results in hypoketonemia
- During fasting, in MCAD deficiency, decreased insulin/ glucagon ratio
- Activation of hormone-sensitive lipase and increased lipolysis in adipose tissue
- Increased free fatty acid levels
- Free fatty acid uptake increases, but, in MCAD deficiency, **β -oxidation defective**
- Low rates of β -oxidation, impaired gluconeogenesis → **severe hypoglycemia**
- Low β -oxidation → **Acetyl CoA NOT formed**
- Liver **ketogenesis impaired** → **hypoketonemia**
- Hypoketonemia also in primary carnitine deficiency, systemic CPT-I deficiency and Jamaican vomiting sickness (Systemic fatty acid oxidation disorders)

Ketogenic diets

- High fat, low carbohydrate diets (Atkins diet) induce **ketosis** – weight loss diets
- Less than 130g/day carbohydrate results in ketosis (Ketogenic diets have 20-40g carbs)
- Low carbohydrate diets are ketogenic
- Management of refractory epilepsy
- Ketogenic diet: Fatty foods such as butter, cream, and peanut butter.
- Limit: Bread, pasta, fruits, and vegetables

Hormone sensitive lipase vs Lipoprotein lipase vs Pancreatic lipase

Hormone sensitive lipase

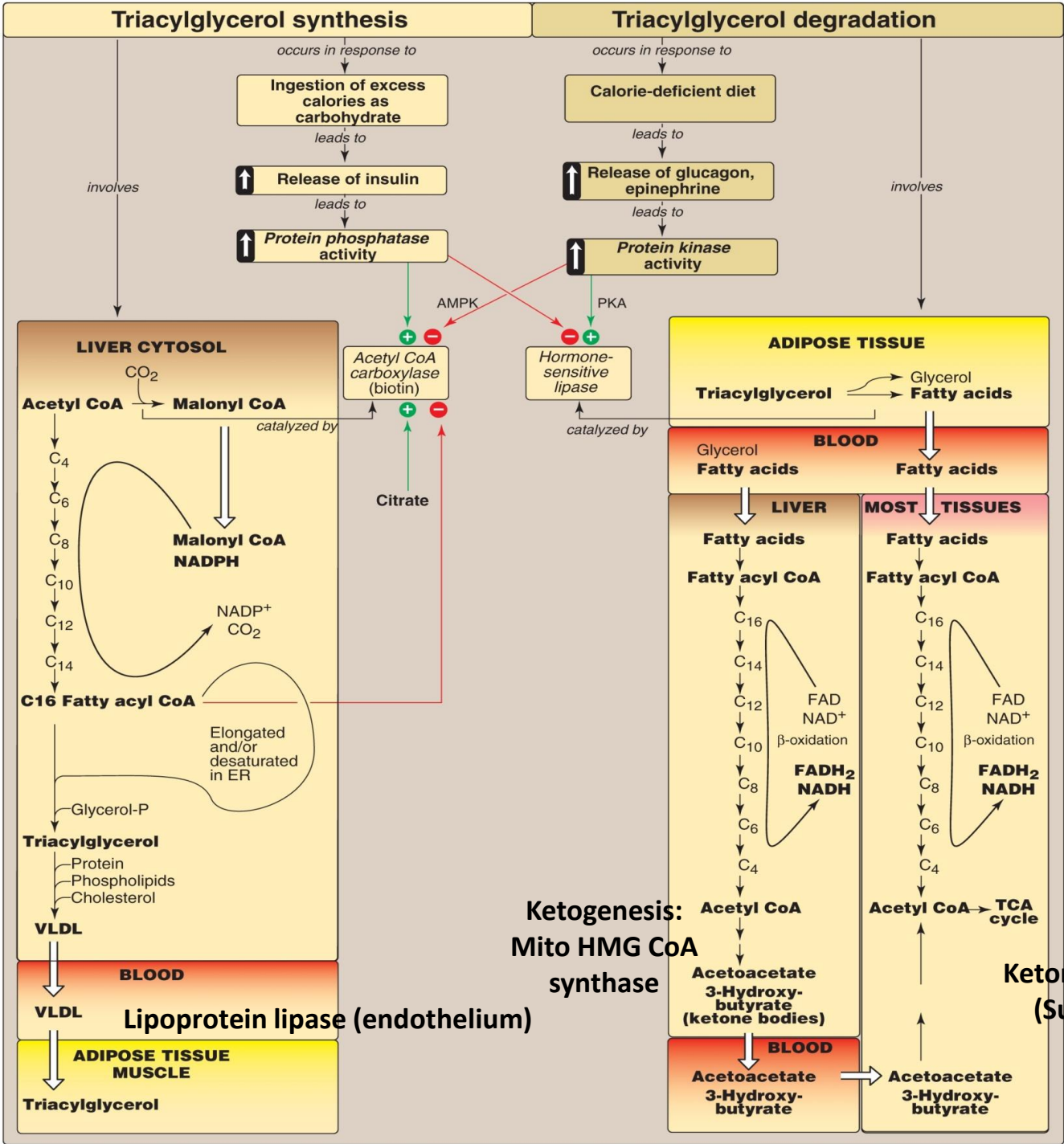
- Location: Within adipocytes
- Role of insulin: Inhibits HS lipase
- Action: Breaks down storage TAG in adipose tissue to free fatty acids and glycerol
- Increased activity of HS lipase:
 - Starvation
 - Insulin resistance (high circulating free fatty acid levels)
 - Uncontrolled type 1 diabetes mellitus

Lipoprotein lipase

- Location: Endothelium of blood vessels in adipose tissue and muscle
- Role of insulin: Induces lipoprotein lipase and required for optimum activity
- Action: Breaks down TAG in chylomicrons and VLDL to free fatty acids that enter adipocytes to be converted to TAG for storage within adipocytes
- Decreased activity of lipoprotein lipase: Metabolic syndrome and Type II diabetes mellitus → Increased serum TAG (VLDL) → increases risk for cardiovascular disease (macrovascular complications)

Pancreatic lipase: Enzyme for digestion of dietary TAG

Big picture: Fatty acid and TAG metabolism



Hormone sensitive lipase (adipocyte)

- Fed state: Fatty acid synthesis in liver
- Fed state: TAG synthesis in liver
- TAG transport as VLDL
- TAG storage in adipose